ARNAUD MERCIER

Computational Geophysicist
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Doctoral researcher at ETH Zurich developing computational methods for efficient full-waveform inversion, with a focus on experimental design, data selection, and differentiable wave simulation.

Education

PhD in Computational Geophysics · Feb 2023–Present
ETH Zurich, Mathematical Geophysics | Zurich, Switzerland
Supervisors: Prof. Hansruedi Maurer, Dr. Christian Boehm
Expected completion: February 2027.

MSc in Earth Sciences and Geophysics (Graduated with Honors) · 2020–2022
Université Laval and INRS, Centre Eau Terre Environnement | Québec, Canada
Supervisors: Prof. J. Christian Dupuis, Prof. Bernard Giroux
GPA: 4.0/4.0 (US scale)

Bachelor of Engineering Physics (Major in Biophotonics) · 2016–2020
Université Laval | Québec, Canada · GPA: 3.9/4.0 (US scale)

Doctoral Research

Doctoral researcher · Feb 2023–Present

ETH Zurich, Mathematical Geophysics | Zurich, Switzerland

Research focus: full-waveform inversion, experimental design and data selection for efficient and robust geophysical inverse problems.

- Developed joint source selection and wavelet model compression for FWI, achieving approximately 50% fewer sources and 90% fewer model parameters in synthetic acoustic studies (GJI, 2025).
- Developed rank-revealing QR data selection for elastic FWI to reduce linear-system size or forward/adjoint simulation counts; evaluated crosshole and surface geometries (preprint, under review).
- Applying data selection to a real near-surface 3D survey for CO_2 storage monitoring, with an HPC workflow designed to avoid assembling the full Jacobian (ongoing).
- Developed a differentiable acoustic wave solver in JAX to investigate receiver placement through full-waveform inversion (paper accepted at the Sim2Science workshop at NeurIPS).
- Implemented research methods in Python and shared code through MODJO; collaborating with startup Mondaic AG researchers on an upcoming open-source package.

Earlier Research Experience

Master's research in geophysical instrumentation · 2020–2022

Université Laval, Geophysical Instrumentation Group | Québec, Canada

Research focus: geophysical instrumentation and vertical seismic profiling.

- Designed seismic sensor test benches, acquisition/communication circuit boards, and microcontroller libraries; processed signals using spectral analysis, filtering, and deconvolution.
- Developed the Geophysical Open Seismic Hardware vertical profiling instrument and contributed to its hardware/firmware repository on GitHub; published the design in the Journal of Open Hardware (2023).

Scientific software and microscopy research · 2018–2019

CERVO Research Center, DCC Lab | Québec, Canada

- Developed processing and visualization software for 200 GB–2 TB microscopy datasets, with unit tests and design documentation (2018).
- Built CNN-based biological image segmentation software, a simulator based on real samples, and a Qt Designer interface (2018–2019).

Peer-reviewed Journal Articles

Mercier, A., Boehm, C., & Maurer, H. (2025). Designing full waveform inverse problems: a combined data and model approach. Geophysical Journal International, 241(3), 1479–1494. 10.1093/gji/ggaf117

Mercier, A., Dupuis, J. C., & Giroux, B. (2023). Geophysical Open Seismic Hardware: Design of a Vertical Seismic Profiling Instrument. Journal of Open Hardware, 7(1), 3. 10.5334/joh.50

Workshop Papers

Mercier, A., Thomsen, H. R., & Moseley, B. (2026). Differentiable Physics for Sensor Placement in Full-Waveform Inversion. Sim2Science: ML with Imperfect Scientific Models, NeurIPS workshop. Accepted. OpenReview

Preprints Under Review

Behnen, K., Maurer, H., **Mercier, A.**, Hertrich, M., & Giardini, D. (2026). Combined Inversion of Active and Passive Seismic Data Based on Optimized Data Selection. Research Square. Preprint; under review. 10.21203/rs.3.rs-10473866/v1

Mercier, A., Boehm, C., & Maurer, H. (2026). Efficient Full-Waveform Inversion via QR-Based Data Selection. EarthArXiv. Preprint; under review. 10.31223/X5PJ4N

Conference Presentations

Mercier, A. (2026). Building Sustainable Scientific Software: Lessons from Industry–Academia. Swiss RSE Day, ETH Zurich, 31 August 2026. Oral presentation.

Mercier, A., & Maurer, H. (2024). Optimizing Full Waveform Inverse Problems: A Combined Data and Model Approach. EGU General Assembly, Vienna, Austria, 14–19 April 2024. Oral presentation, EGU24-17729.

Mercier, A., Dupuis, J. C., & Giroux, B. (2022). Open hardware: Geophysical instrumentation for collaborations in a changing world. IMAGE, Houston, USA. Oral presentation.

Mercier, A. (2022). Open seismic hardware: a step towards accessibility in mineral exploration. GeoConvention, Calgary, Canada. Oral presentation.

Teaching Experience

Numerical Methods for Geophysics in Master of Geophysics at ETH Zurich · 2023–2026

- Designed and led practical sessions on numerical methods for geophysics.
- Designed and graded examinations.

Reflection Seismic Processing in Master of Geophysics at ETH Zurich · 2023–2026

- Supported students in reflection seismic processing, helping them break down a complex, multistep workflow into clear, manageable stages.

Geophysical methods in Geology bachelor at Laval University · 2021–2023

- Led tutorials and Q&A sessions, graded exercises and examinations, and provided individual support for Python programming exercises.

Thesis Supervision

ETH Zurich, Mathematical Geophysics · 2025–2026

Master's thesis: Experimental Characterisation of Elastic Guided Waves in Thin Plates: Toward Full-Waveform Inversion. (Main supervisor)

Bachelor's thesis: Lake-Ice Thickness Mapping Using Simulated Seismo-Acoustic Signals. (Co-supervisor)

Fieldwork Experience

3D land seismic survey (2025): Led the completion of a survey comprising 1,000 seismic nodes and 150 shots; coordinated a four-person team and liaised with contractors.

Avalanche monitoring (2024): Deployed a fiber-optic cable on an avalanche-prone slope; supported fiber layout and anchoring system design.

Reviewer Activities

Applied Computing and Geosciences, Journal of Petroleum Exploration and Production Technology, NeurIPS Workshop.

Technical Skills and Languages

Scientific computing: Python, JAX, Optax, MATLAB, C++, C; Linux; Git CI/CD.

Numerical methods: Wave simulation, automatic differentiation, non-linear optimization, QR factorization, wavelets, experimental design.

Machine learning and data: CNN image segmentation, volumetric imaging, seismic signal processing.

Instrumentation: Sensor testing, PCB design, microcontrollers; Altium, SolidWorks, Creo, AutoCAD.

Languages: French (native), English (native) and German (proficient).

Academic Leadership and Awards

Co-founder, designer and coordinator, REDCO, Université Laval. Coordinated 15 members across four optics projects; developed fiber-optic sensor prototypes and test benches (2017–2020).

Winner, “My thesis in 5 minutes”, Université Laval geology and mineral resources research center colloquium (2021).

Admission to the Autumn School on Scientific Machine Learning and Numerical Methods at CWI in Amsterdam (2025)